

TrES-2b

R-2014

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_121025 · created 2026-08-07T23:19:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456225.65827 +/- 0.00018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.02
Transit depth (Rp/Rs)^2	1.14 +/- 0.42 [%]
Orbital Inclination (inc)	83.82 +/- 0.65
Airmass coefficient 1 (a1)	1.2426 +/- 0.009
Airmass coefficient 2 (a2)	-0.1389 +/- 0.0037
Scatter in the residuals of the lightcurve fit is	0.72 %
Best Comparison Star	None
Optimal Aperture	4.18
Optimal Annulus	8.67
Transit Duration (day)	0.069 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.107 ± 0.02 vs 0.1254 (deviation 0.92σ)
Duration fit vs published	0.069 d vs 0.0746 d (deviation 0.43σ)
Mid-time O–C	-0.4 min
Depth SNR	5.3
Residual scatter	0.72 %

Light curve

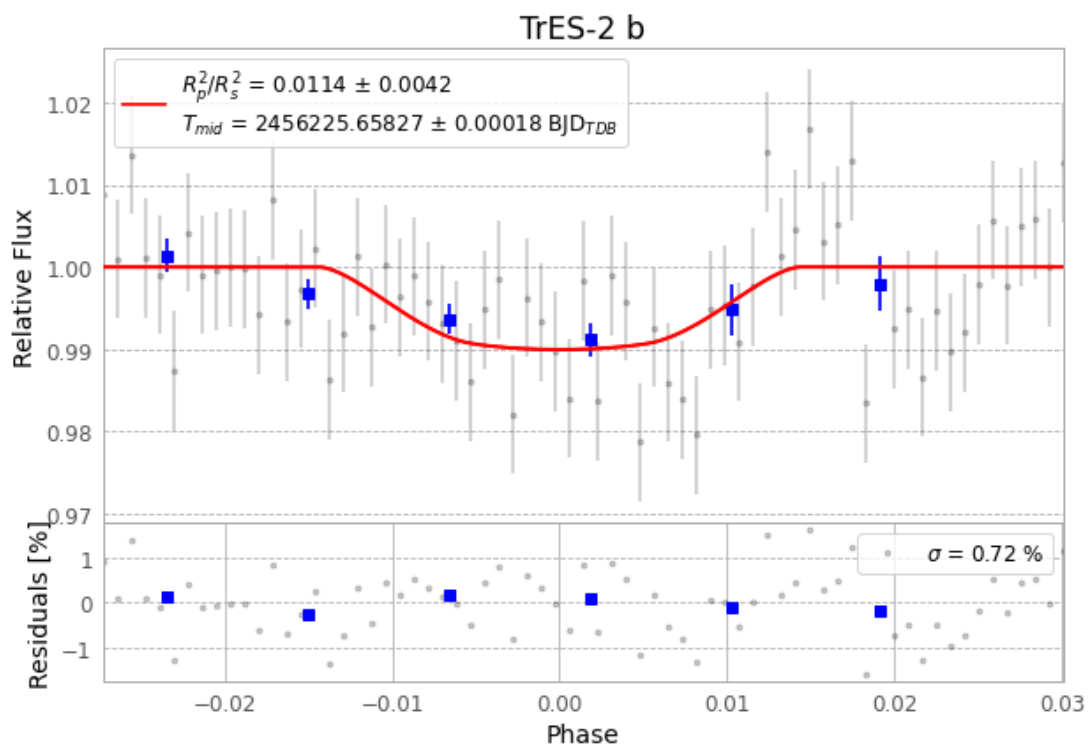


Figure 1 — FinalLightCurve_TrES-2 b_2012-10-24.png (SHA-256 5f1491bed5825f18...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [256, 270], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 680bb8a218383dc7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

69 FITS frames (45 MB), TRES-2121025020912.FITS → TRES-2121025053312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-121025.log (e2baa6d2bffe...), FinalLightCurve_TrES-2 b_2012-10-24.png (5f1491bed582...), FinalLightCurve_TrES-2 b_2012-10-24.csv (e3fa620c822a...)

Compute resources

Wall time 131.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 23:19Z.